

Stock Price Predictor

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Part 1

Introduction

Research Question

Can we predict whether a stock's closing price will be higher or lower than its opening price on a given day?

Our Motivation

Reasons:

- Direct Real-World Application
- Learning Opportunities
- Modifiable Complexity

Notable Models:

- Ensemble DL
- ResNLS
- StockTime



Part 2

Data

Data

- **Data Source:** Historical financial data from Yahoo Finance focusing on S&P 500 companies
- **Time Frame:** First Model: data back to 2000, Second Model: 1/1/21 - 1/1/25 (4 years), Current Model: 1/1/21 - 3/26/25
- **Dataset Composition:** ~500 stocks with 5 key daily features: Open, High, Low, Close, and Volume
- **Ticker Sourcing:** Initial ticker list from Wikipedia; eliminated survivorship bias
- **Data Cleaning:** Removed duplicates and nans and restructured data through unstacking
- **Summary Statistics:** Computed metrics such as average price and volume
 - Note: these averages over all 500 stocks are of limited relevance
- **Size:** 478,380 rows and 8 columns

index	Open	High	Low	Close	Volume
mean	166.640	168.517	164.734	166.646	5954927
std	350.228	354.214	346.395	350.333	2393987
min	1.00999	1.03999	0.77999	0.98000	0.0
25%	50.7755	51.3660	50.1565	50.7700	999575
50%	97.6452	98.8331	96.4500	97.6490	2048900
75%	187.429	189.466	185.368	187.390	4708725
max	9914.16	9964.76	9794.0	9924.40	1543911

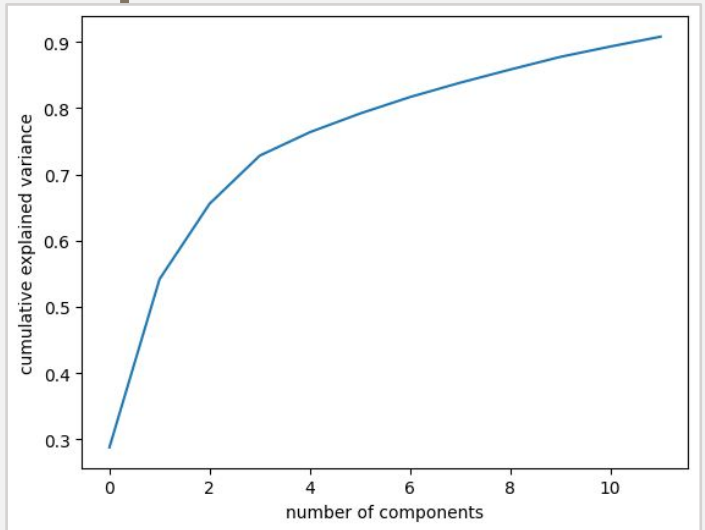
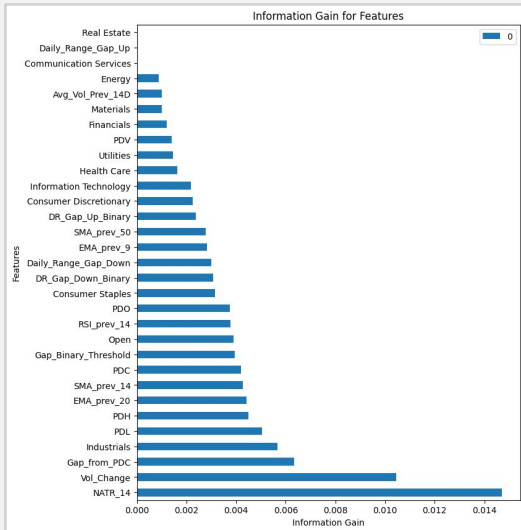
Features

- Most possible features can be derived from the Open, High, Low, Close, and Volume metrics
 - a. From both current day and previous day prices
- Theoretically infinite number of derived features, but ultimately landed on well-known metrics and logical features
- **Caveat: cannot use features from the current day except for Open price**
 - a. Emphasis on previous day metrics
 - i. PDO: Previous day open (price)
 - ii. PDH: Previous day high (price)
 - iii. PDL: Previous day low (price)
 - iv. PDC: Previous day close (price)
 - v. PDV: Previous day volume

Target Variable: stock performance relative to open price

- 2: Long Case
- 1: Neutral Case
- 0: Short Case

PCA/Feature Importance



Part 3

Modeling

Model Timeline

Non-Time Series Models

Initial LSTM Model

LSTM Model with Sliding Windows

Window/Rolling Window Models

LSTM Model with PCA



Model Timeline

Non-Time Series Models

- Linear/Logistic [Binary/Ternary]
- Shuffled Time Series
- Metric: Accuracy, RMSE
- Poor Generalization

Initial LSTM Model

LSTM Model with Sliding Windows

Window/Rolling Window Models

LSTM Model with PCA



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- Linear/Logistic [Binary/Ternary]
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Initial LSTM Model

LSTM Model with Sliding Windows

Window/Rolling Window Models

- Maintain Time Series
- Metric: Accuracy
- Batch Modeling

LSTM Model with PCA



Model Timeline

Non-Time Series Models

- Linear/Logistic [Binary/Ternary]
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- Metric: Accuracy, RMSE
- Poor Generalization

Initial LSTM Models

- Normalize: Open | High
- Train/Test Split: 80/20 | 99/1
- All | AAPL | Sector
- Sequence Length: 14 days, 35 days

LSTM Model with Sliding Windows

Window/Rolling Window Models

- Maintain Time Series
- Metric: Accuracy
- Batch Modeling

LSTM Model with PCA

Model Timeline

Non-Time Series Models

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Initial LSTM Models

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LSTM Model with Sliding Windows

Window/Rolling Window Models

- Maintain Time Series
- Metric: Accuracy
- Batch Modeling

LSTM Model with PCA

- 45 Features -> 13 Features
- Poorer Generalization
- Marginal computation streamlining

Part 4

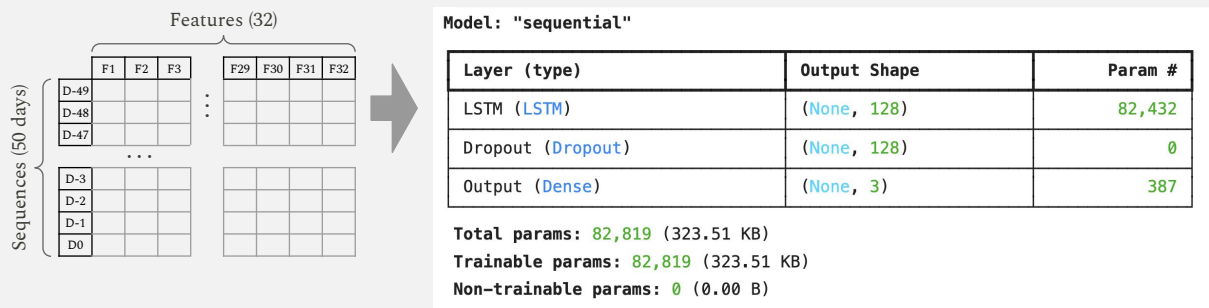
Experiments and Final Model Architecture

LSTM Model Experiments

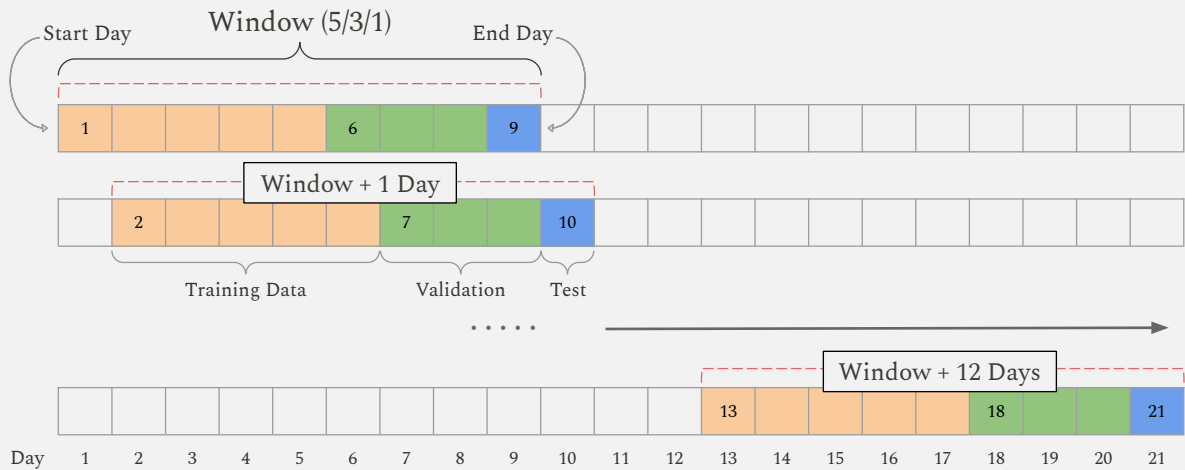
Model	Sequence length	LSTM units	LSTM layers	Train/test split	Features	Learning rate	Optimizer	Batch size	Epochs	Training accuracy*	Val accuracy*	Test accuracy*
Default LSTM Model	14	128	2	80/20	32	0.0001	Adam	1,024	10	0.4080	0.4158	--
Learning-Rate Adjusted Model	14	128	2	80/20	32	0.00001-0.001	SGD/Adam	1,024	10-50	0.4961	0.4903	0.4023
Longer Series Model	35	256	2	80/20	32	0.001	SGD/Adam	1,024	10	0.5537	0.5299	0.4087
Smaller Train/Test Split (99/1) Model	14	256	2	99/1	32	0.001	Adam	256	10	0.5598	0.5159	0.5246
99/1 PCA Model	14	256	2	99/1	13 (comps.)	0.0001-0.001	Adam	256	10	0.5318	0.4802	0.4842
99/1 Longer Series Model	35	256	2	99/1	32	0.001	Adam	1,024	10-15	0.5471	0.5363	0.5265
99/1 LSTM Single-Layer Model	35	256	1	99/1	32	0.001	Adam	256	10-13	0.5651	0.5291	0.5204
1-Day Test Size Model	35	256	1	1 test day	32	0.001	Adam	256	10	0.5429	0.5453	0.54274

*Note: These experiments were done with the incorrect data, so the accuracies are not absolute, and they can only be compared in relation to each other.

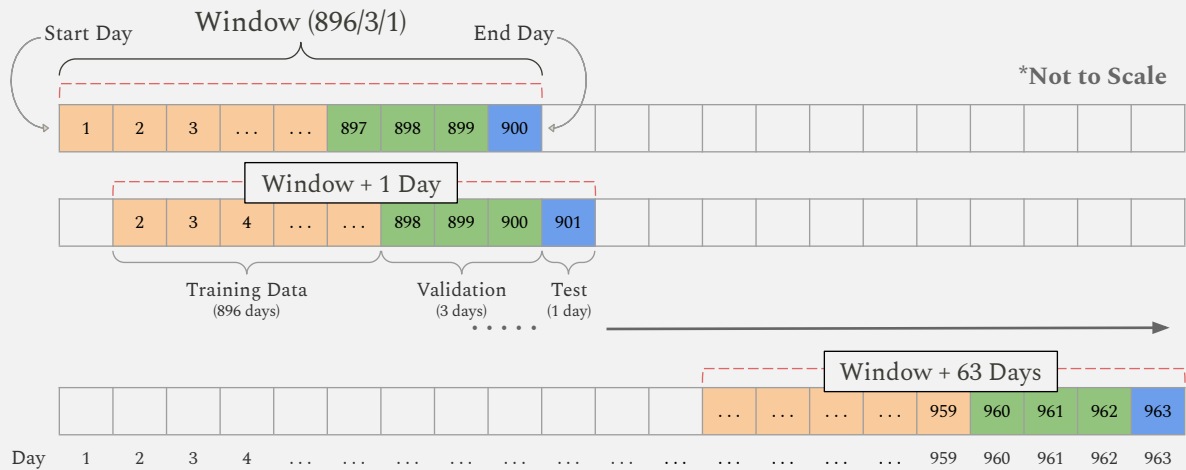
Final LSTM Model Architecture



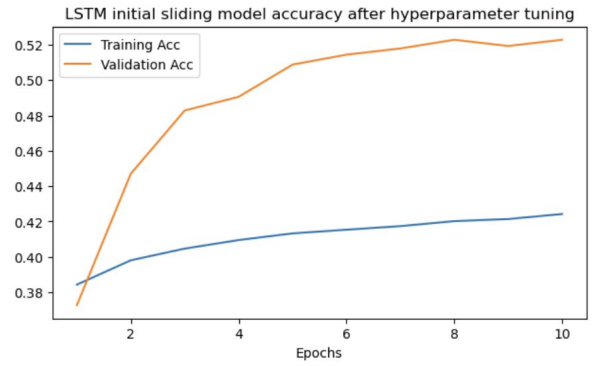
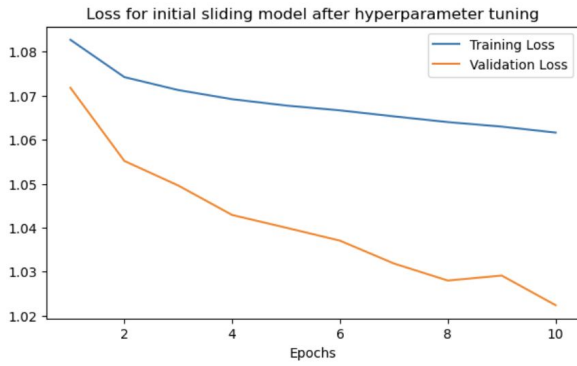
Sliding Window LSTM Model



Sliding Window LSTM Model



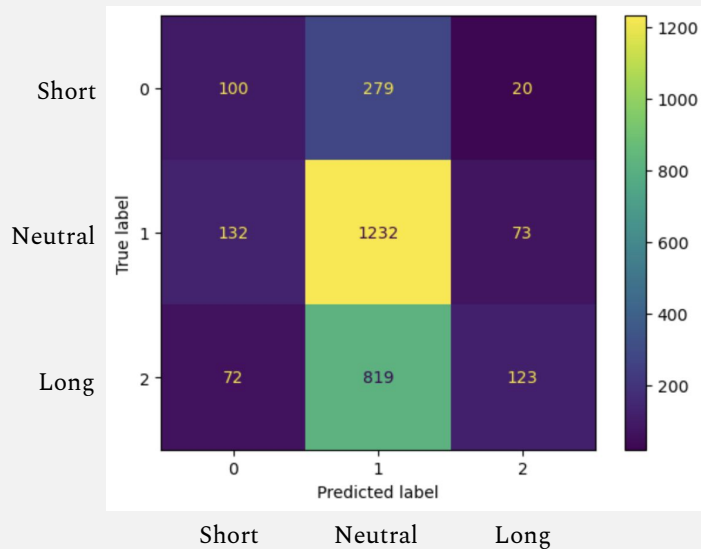
Model Results for 1st Window



Final accuracies:

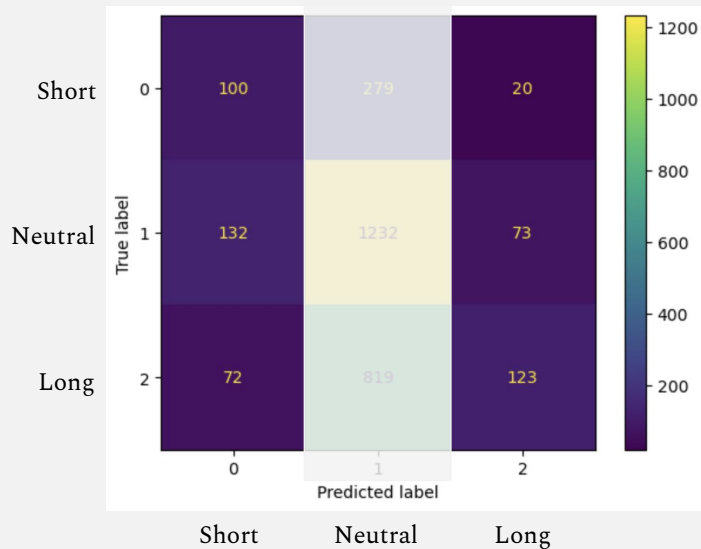
- Training accuracy: 0.4242
- Validation accuracy: 0.5228

Confusion Matrix (6 Test Days)



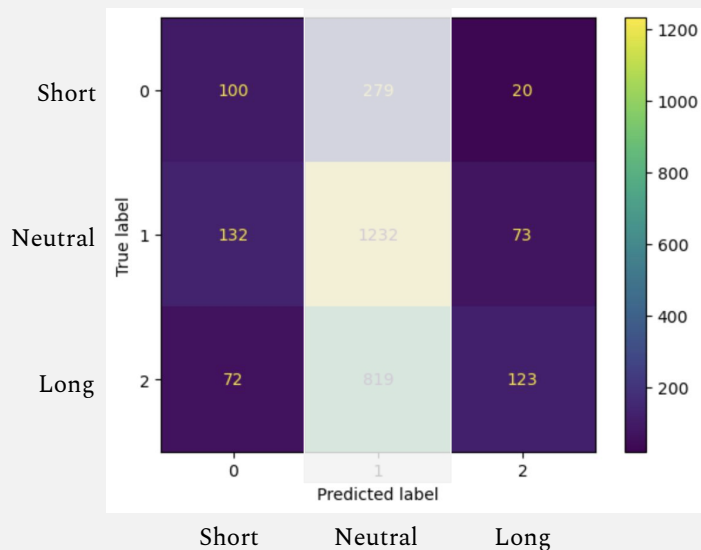
Note: we use a precision threshold of 0.5 for Class 0 and Class 2 (i.e., if the softmax prediction is not at least 0.5 for Class 0 or 2, it will default to Class 1).

Confusion Matrix (6 Test Days)



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Confusion Matrix (6 Test Days)



Precision Formula:

$$\frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

Class 0 (Short) Precision:

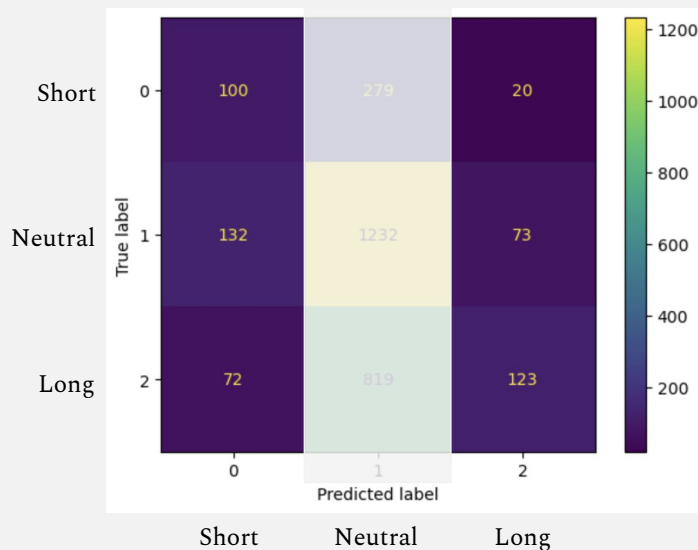
$$\frac{100}{100 + 132 + 72}$$

Class 2 (Long) Precision:

$$\frac{123}{123 + 73 + 20}$$

Note: we use a precision threshold of 0.5 for Class 0 and Class 2 (i.e., if the softmax prediction is not at least 0.5 for Class 0 or 2, it will default to Class 1).

Confusion Matrix (6 Test Days)



Precision Formula:

$$\frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

Class 0 (Short) Precision:

$$\frac{100}{(100 + 132 + 72)} = \sim 33\%$$

Class 2 (Long) Precision:

$$\frac{123}{(123 + 73 + 20)} = \sim 57\%$$

Note: we use a precision threshold of 0.5 for Class 0 and Class 2 (i.e., if the softmax prediction is not at least 0.5 for Class 0 or 2, it will default to Class 1).

Single Day Precisions

Day 42				
0	5	146	3	
1	0	138	1	
2	4	176	2	
	0	1	2	
Day 43				
0	6	230	2	
1	1	158	1	
2	1	72	4	
	0	1	2	
Day 44				
0	4	216	7	
1	1	138	1	
2	2	104	2	
	0	1	2	
Day 45				
0	10	34	0	
1	1	119	3	
2	6	291	11	
	0	1	2	

...

Class 0 Prec: 56%
Class 2 Prec: 33%

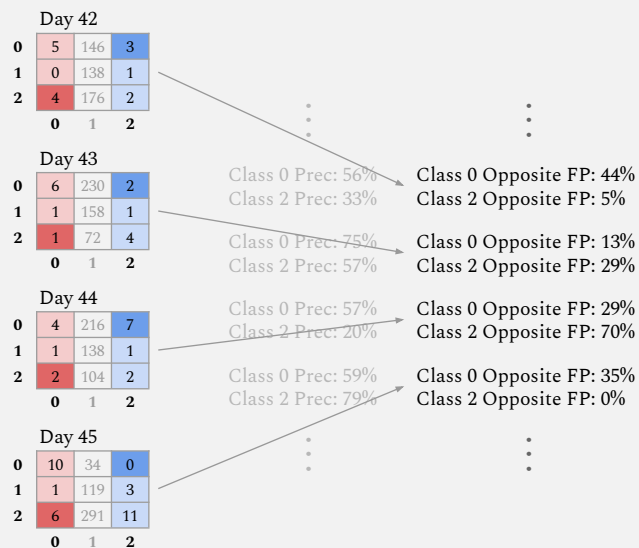
Class 0 Prec: 75%
Class 2 Prec: 57%

Class 0 Prec: 57%
Class 2 Prec: 20%

Class 0 Prec: 59%
Class 2 Prec: 79%

...

Single Day Opposite FPs



Single Day Neutral FPs

Day 42

0	5	146	3
1	0	138	1
2	4	176	2
	0	1	2

⋮

⋮

⋮

Day 43

0	6	230	2
1	1	158	1
2	1	72	4
	0	1	2

Class 0 Prec: 56%
Class 2 Prec: 33%

Class 0 Opposite FP: 44%
Class 2 Opposite FP: 5%

Class 0 Neutral FP: 0%
Class 2 Neutral FP: 17%

Class 0 Prec: 75%
Class 2 Prec: 57%

Class 0 Opposite FP: 13%
Class 2 Opposite FP: 29%

Class 0 Neutral FP: 13%
Class 2 Neutral FP: 14%

Day 44

0	4	216	7
1	1	138	1
2	2	104	2
	0	1	2

Class 0 Prec: 57%
Class 2 Prec: 20%

Class 0 Opposite FP: 29%
Class 2 Opposite FP: 70%

Class 0 Neutral FP: 14%
Class 2 Neutral FP: 10%

Class 0 Prec: 59%
Class 2 Prec: 79%

Class 0 Opposite FP: 35%
Class 2 Opposite FP: 0%

Class 0 Neutral FP: 6%
Class 2 Neutral FP: 21%

Day 45

0	10	34	0
1	1	119	3
2	6	291	11
	0	1	2

⋮

⋮

⋮

Average Precisions and FPs

Day 42

0	5	146	3
1	0	138	1
2	4	176	2
	0	1	2

⋮

⋮

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Day 45

0	10	34	0
1	1	119	3
2	6	291	11
	0	1	2

⋮

⋮

⋮

Average Precisions and False Positive Rates Across 64 Test Days

Average precisions:

- Class 0: 53%
- Class 2: 55%

Average opposite FP rate:

- Class 0: 28%
- Class 2: 31%

Average neutral FP rate:

- Class 0: 19%
- Class 2: 14%

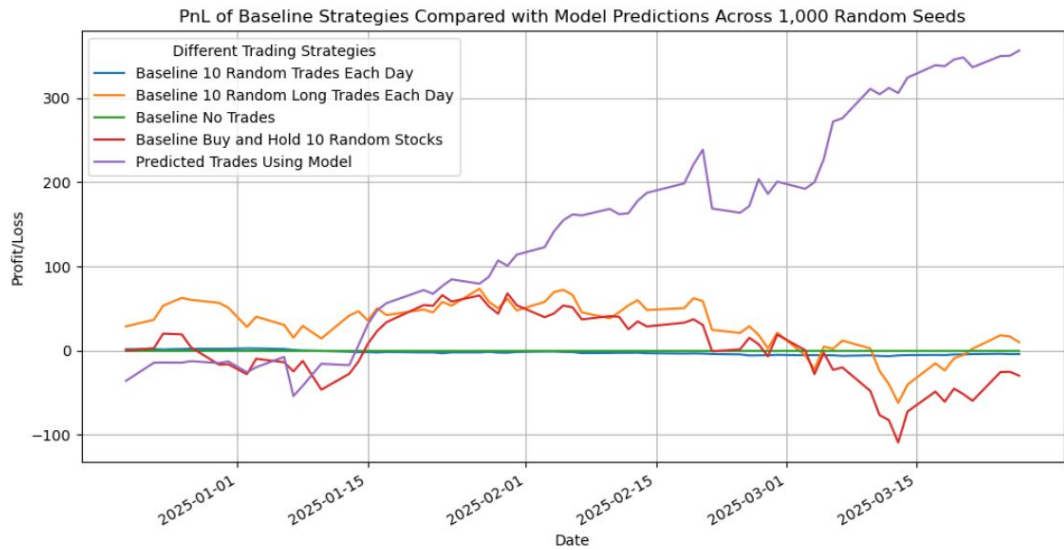
Part 5

Conclusion

Conclusion

Label	Earn Money	Break Even	Lose Money
Sell Short (Class 0)	~53% Chance	~19% Chance	~28% Chance
Buy Long (Class 2)	~55% Chance	~14% Chance	~31% Chance

Real-World Modeling



Challenges + Future Work

- Challenges
 - Nature of Stock Data
 - High Computational Cost
- Future Work
 - Functional API
 - Embeddings
 - Sector-Based
 - More Features
 - Alternative architectures
 - GRUs & Transformers

**THANK
YOU!**

Appendix

Feature Selection

Gap from PDC

- $(\text{Open Price} - \text{PDC Price}) / \text{PDC Price}$

Gap Binary Threshold

- $\text{abs}(\text{Gap_from_PDC}) > 0.02$

Daily Range Gap Up Binary

- $(\text{Open Price} - \text{PDH}) / \text{PDH} > 0$

Daily Range Gap Down Binary

- $(\text{Open Price} - \text{PDL}) / \text{PDL} < 0$

Volume Change

- $(\text{Volume} - \text{Previous 14-Day Avg Vol}) / \text{Previous 14-Day Avg Vol}$

RSI Previous 14 Days

$$RSI_{\text{step two}} = 100 - \left[\frac{100}{1 + \left(\frac{(\text{Previous Average Gain} \times 13) + \text{Current Gain}}{((\text{Previous Average Loss} \times 13) + \text{Current Loss})} \right)} \right]$$

[44]:	Gap_from_PDC	Gap_Binary_Threshold	DR_Gap_Up_Binary	DR_Gap_Down_Binary	Vol_Change	RSI_prev_14
Date						
2021-01-04	NaN	False	False	False	NaN	NaN
2021-01-05	-0.004018	False	False	False	NaN	NaN
2021-01-06	-0.025113	True	False	True	NaN	NaN
2021-01-07	0.013902	False	False	False	NaN	NaN
2021-01-08	0.011533	False	True	False	NaN	NaN
...	...	...	...	...	...	...
2024-12-24	0.000862	False	False	False	-0.550122	73.085634
2024-12-26	-0.000039	False	False	False	-0.456735	75.750267
2024-12-27	-0.004594	False	False	False	-0.139503	76.452855
2024-12-30	-0.013146	False	False	True	-0.283312	67.626289
2024-12-31	0.000952	False	False	False	-0.193683	60.225595

Feature Selection

**Global Industry Classification
Standard (GICS) Sector**

- Indicates the industry a stock is in

DR_Gap_Down_Binary	Vol_Change	RSI_prev_14	GICS Sector
False	-0.172814	53.257057	Health Care
False	-0.329771	61.017822	Financials
False	0.348177	63.870569	Energy
False	-0.544614	57.145044	Health Care
False	-0.229278	44.530455	Information Technology
...	...	...	...
False	-0.215393	57.267100	Real Estate
False	0.355798	47.729828	Information Technology
False	0.012974	42.397260	Industrials
True	0.080185	58.185330	Communication Services
True	-0.329750	58.807894	Communication Services

Baseline Models

No Time Series (Logistic Regression Classification & Linear Regression Models)

Ternary Logistic Regression Model (Single-Ticker/Multi-Ticker)

- Multinomial Logistic Regression (with/without hidden layers)

Linear Regression Model (Single-Ticker/Multi-Ticker)

- Linear Model (with/without hidden layers)

Attribute(s):

- All tickers and sector feature
- Multiple hidden layers
- Tuned hyperparameters via Keras Tuner
- Differing batch sizes and optimizers
- Metric: Accuracy
- Both Logistic & Linear Regression Models

Conclusion:

Poor generalization

- Model assumes each observation is independent of the other observations

Baseline Models cont.

Time Series OHLCV (Rolling Windows: 14 Days)

Ternary Logistic Regression Model with Rolling Window (Single-Ticker/Multi-Ticker)

- Multinomial Logistic Regression (with hidden layers)

Attribute(s):

- Open | High | Low | Close | Volume Features
- 14 Day Rolling Window (pseudo-time-series)

Conclusion:

Poor generalization

- Model assumes each observation is independent of the other observations

LSTM Model

LSTM:

- Models: 80/20 day, 99/1 day, Sliding Window (100 day | 700 day)
- All Stocks (14 day sequence), AAPL, Sector, All Stocks (35 day sequence)
- Normalization: Open | Max value for each Ticker (Min val: 0 - Max val: 2)
 - Preservation of long-term price changes
 - Ideally mean value lands at ~1 to match other features that are scaled differently
- Ternary Classification: Positive Prediction | Neutral Prediction | Negative Prediction

Conclusion:

Poor generalization

- Model assumes each observation is independent of the other observations

Final LSTM Model Architecture

Model: "sequential"

Layer (type)	Output Shape	Param #
LSTM (LSTM)	(None, 128)	82,432
Dropout (Dropout)	(None, 128)	0
Output (Dense)	(None, 3)	387

Total params: 82,819 (323.51 KB)

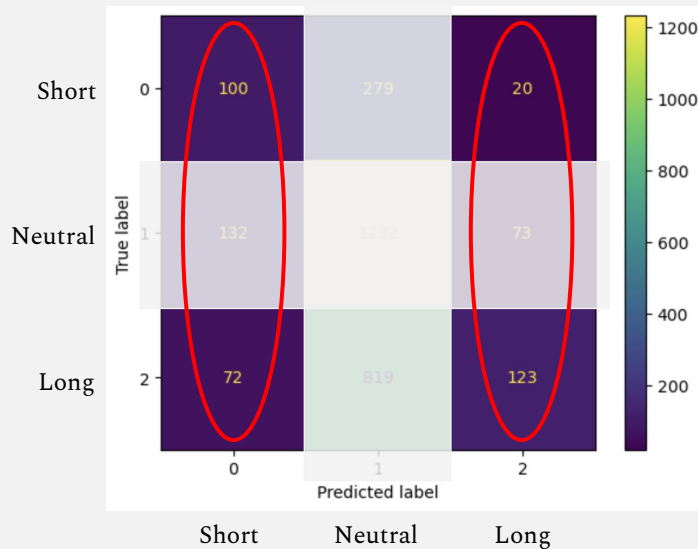
Trainable params: 82,819 (323.51 KB)

Non-trainable params: 0 (0.00 B)

LSTM Sliding Window Model Parameters:

- Sequence length: 50 (days)
- Features: 32
- Train/val/test split: 896/3/1
- LSTM units: 128
- LSTM layers: 1
- Dropout rate: 0.3
- Optimizer: Adam
- Epochs: 10
- Learning rate: 0.0001
- Batch size: 256

Confusion Matrix (6 Test Days)



Ratio between True Positives and “Opposite Class” False Positives

Class 0:

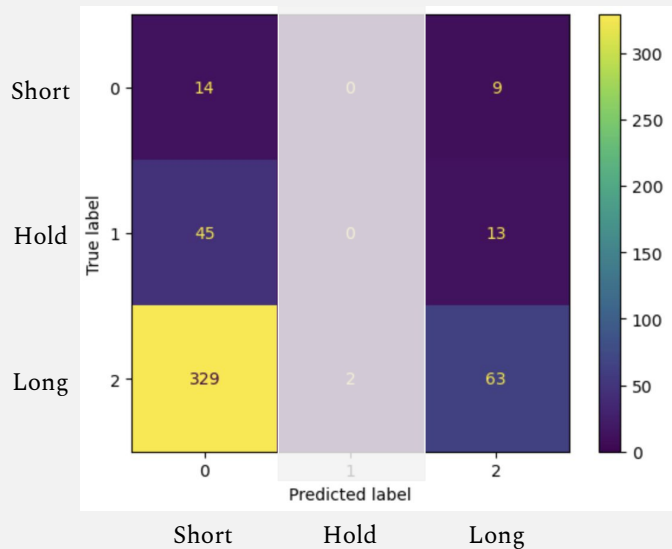
$$100:72 = \sim 1.39$$

Class 2:

$$123:20 = 6.15$$

Note: we use a precision threshold of 0.5 for Class 0 and Class 2 (i.e., if the softmax prediction is not at least 0.5 for Class 0 or 2, it will default to Class 1).

Confusion Matrix (1 Test Day)



Precision Formula:

$$\frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

Class 0 (Short) Precision:

$$\frac{14}{(14 + 45 + 329)} = \sim 3.6\%$$

Class 2 (Long) Precision:

$$\frac{63}{(63 + 13 + 9)} = \sim 74\%$$

Note: no prediction threshold was used here.

Average Precisions and FPs

Day 42

0	5	146	3
1	0	138	1
2	4	176	2
	0	1	2

⋮

⋮

⋮

Day 43

0	6	230	2
1	1	158	1
2	1	72	4
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Class 0 Prec: 56%
Class 2 Prec: 33%

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Class 2 Opposite FP: 5%

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0	4	216	7
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Class 0 Prec: 57%
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Class 2 Opposite FP: 70%

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Day 45

0	10	34	0
1	1	119	3
2	6	291	11
	0	1	2

⋮

⋮

⋮

**Average Precisions and
False Positive Rates
Across 64 Test Days**

Average precisions:

- Class 0: 53%
- Class 2: 55%

Average opposite FP rate:

- Class 0: 28%
- Class 2: 31%

Average neutral FP rate:

- Class 0: 19%
- Class 2: 14%

General Model Results

**See note below for caveat*

Note: While the relative accuracies shown in the following slides is representative of how each model performs, the absolute accuracies are **incorrect** due to a data processing error. To change the data for and re-run all the models shown below would have taken too much time, so we chose to keep what we had and move on with the correct data. This isn't to say, however, that our results are worthless, as we saw how altering the different hyperparameters, sizes, sequence length, model complexity, etc. affected model performance, and it ultimately led us to the final model we ended up with.

Initial LSTM Model Architecture

Layer (type)	Output Shape	Param #
LSTM_1 (LSTM)	(None, 14, 128)	82,432
Dropout_1 (Dropout)	(None, 14, 128)	0
LSTM_2 (LSTM)	(None, 128)	131,584
Dropout_2 (Dropout)	(None, 128)	0
Output (Dense)	(None, 3)	387

Total params: 214,403 (837.51 KB)

Trainable params: 214,403 (837.51 KB)

Non-trainable params: 0 (0.00 B)

Sequence Length (Days)

LSTM Units

Consistent LSTM Model Parameters:

- Classes: 3
- Optimizer: Adam
- Epochs: 10
- Features: 32 (until PCA)

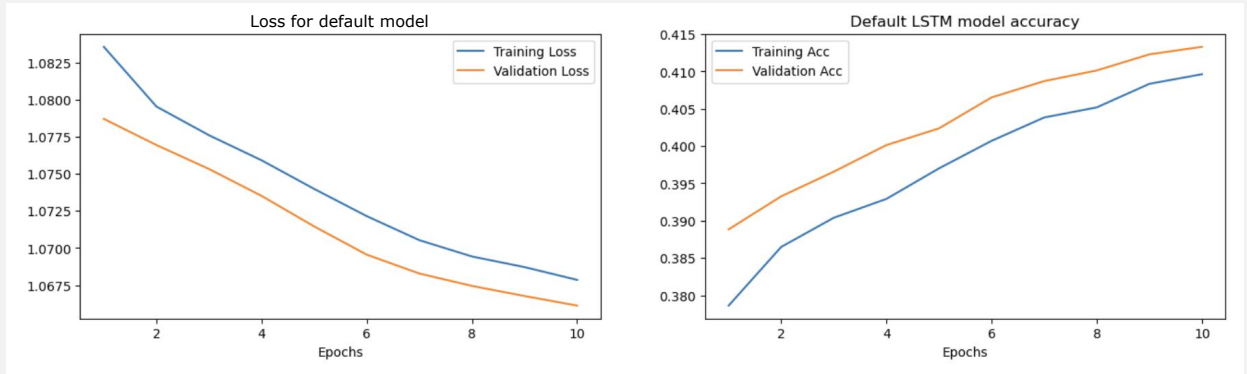
Default LSTM Model

Hyperparameters:

- Sequence length: 14 (days)
- LSTM units: 128
- LSTM layers: 2
- Train/test split: 80/20
- Learning rate: 0.0001
- Batch size: 1,024

Final accuracies*:

- Training accuracy: 0.4080
- Validation accuracy: 0.4158



*Pay not much attention to the absolute accuracies, but look at the relative accuracies between them

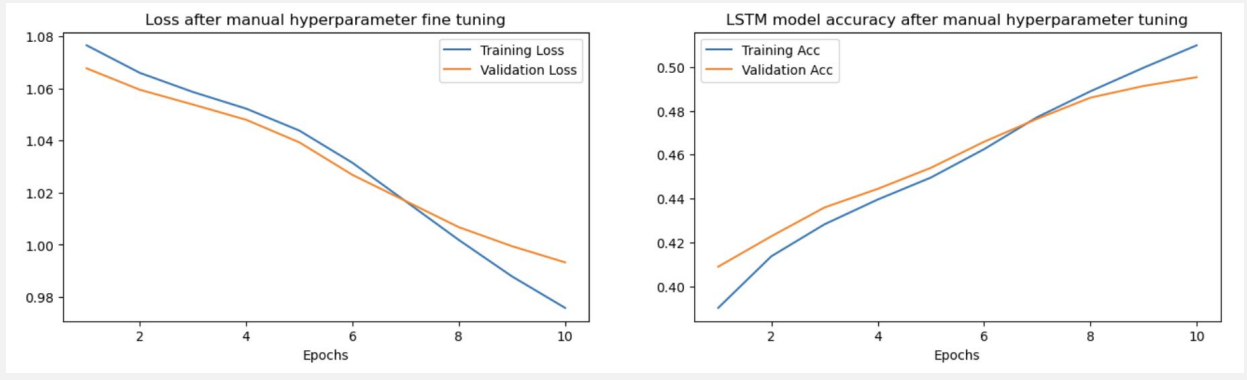
Learning-Rate Adjusted Model

Hyperparameters:

- Sequence length: 14 (days)
- LSTM units: 128
- LSTM layers: 2
- Train/test split: 80/20
- Learning rate: 0.001
- Batch size: 1,024

Final accuracies*:

- Training accuracy: 0.4961
- Validation accuracy: 0.4903
- Test accuracy: 0.4023



*Pay not much attention to the absolute accuracies, but look at the relative accuracies between them

Starting with this slide, every new model shown is after hyperparameter-tuning to the best parameters

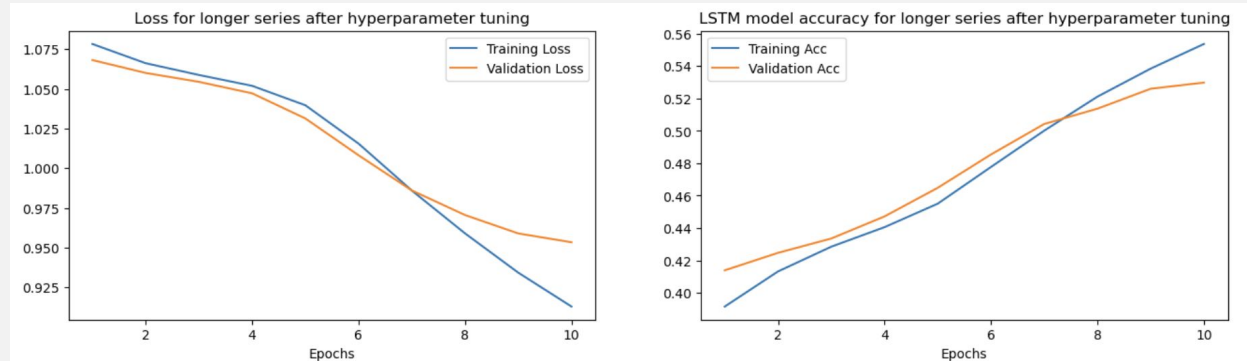
With Longer Series

Hyperparameters:

- Sequence length: 35 (days)
- LSTM units: 256
- LSTM layers: 2
- Train/test split: 80/20
- Learning rate: 0.001
- Batch size: 1,024

Final accuracies*:

- Training accuracy: 0.5537
- Validation accuracy: 0.5299
- Test accuracy: 0.4087



*Pay not much attention to the absolute accuracies, but look at the relative accuracies between them

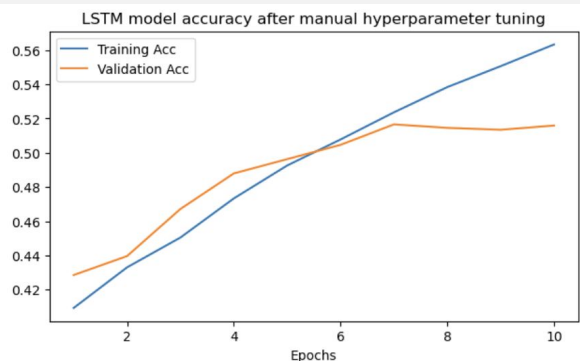
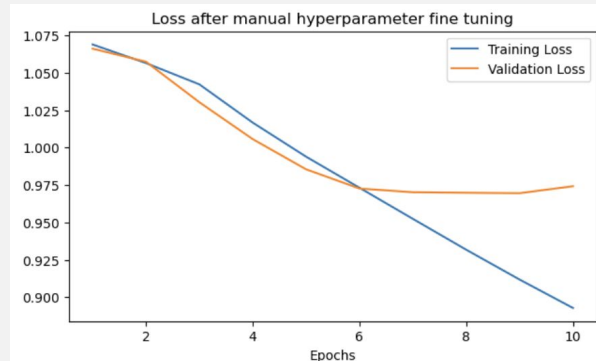
Smaller Test/Val Sizes

Hyperparameters:

- Sequence length: 14 (days)
- LSTM units: 256
- LSTM layers: 2
- Train/test split: 99/1
- Learning rate: 0.001
- Batch size: 256

Final accuracies*:

- Training accuracy: 0.5598
- Validation accuracy: 0.5159
- Test accuracy: 0.5246



*Pay not much attention to the absolute accuracies, but look at the relative accuracies between them

With PCA

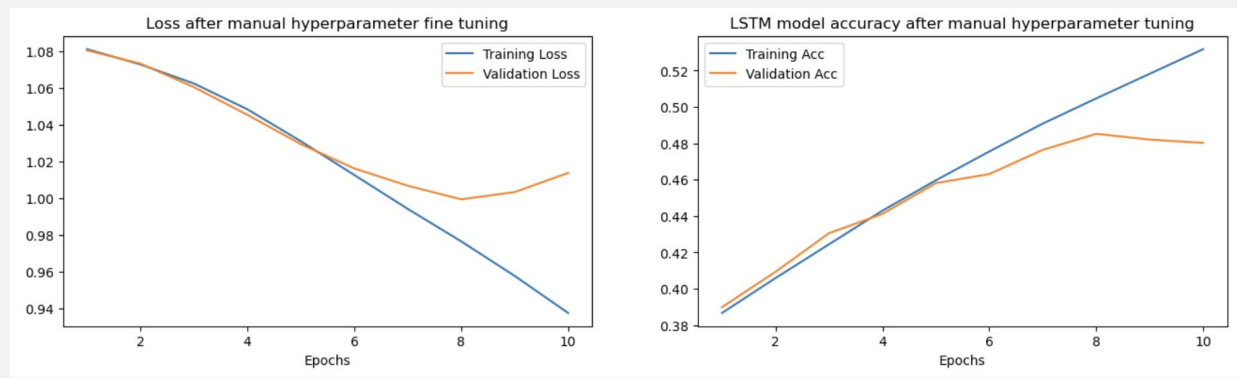
Hyperparameters:

- Sequence length: 14 (days)
- LSTM units: 256
- LSTM layers: 2
- Train/test split: 99/1
- Learning rate: 0.001
- Batch size: 256

- **Components: 13**

Final accuracies*:

- Training accuracy: 0.5318
- Validation accuracy: 0.4802
- Test accuracy: 0.4842



*Pay not much attention to the absolute accuracies, but look at the relative accuracies between them

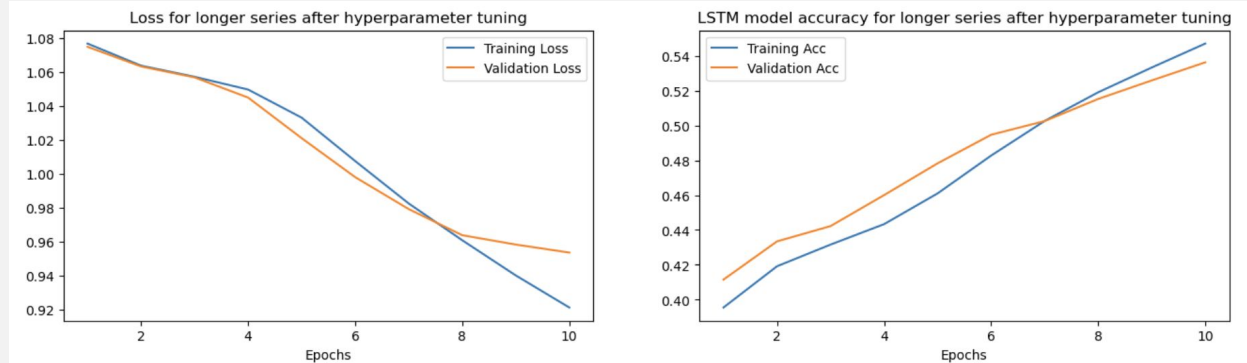
With Longer Series

Hyperparameters:

- Sequence length: 35 (days)
- LSTM units: 256
- LSTM layers: 2
- Train/test split: 99/1
- Learning rate: 0.001
- Batch size: 1024

Final accuracies*:

- Training accuracy: 0.5471
- Validation accuracy: 0.5363
- Test accuracy: 0.5265



*Pay not much attention to the absolute accuracies, but look at the relative accuracies between them

Using Simpler LSTM Model

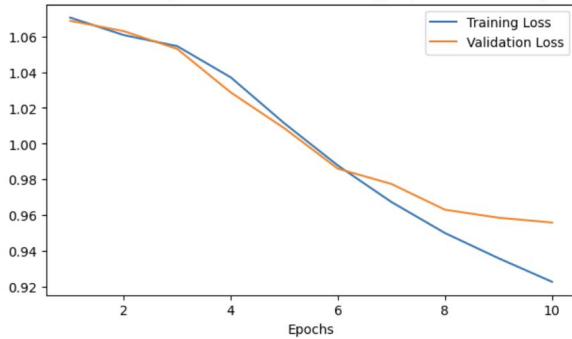
Hyperparameters:

- Sequence length: 35 (days)
- LSTM units: 256
- LSTM layers: 1
- Train/test split: 99/1
- Learning rate: 0.001
- Batch size: 256

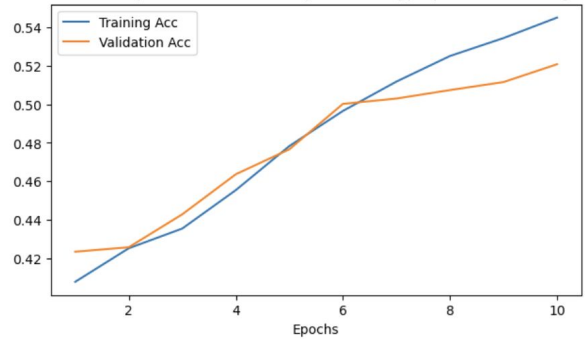
Final accuracies*:

- Training accuracy: 0.5651
- Validation accuracy: 0.5291
- Test accuracy: 0.5204

Loss for simplified LSTM model after hyperparameter tuning



LSTM simplified model accuracy for after hyperparameter tuning



*Pay not much attention to the absolute accuracies, but look at the relative accuracies between them

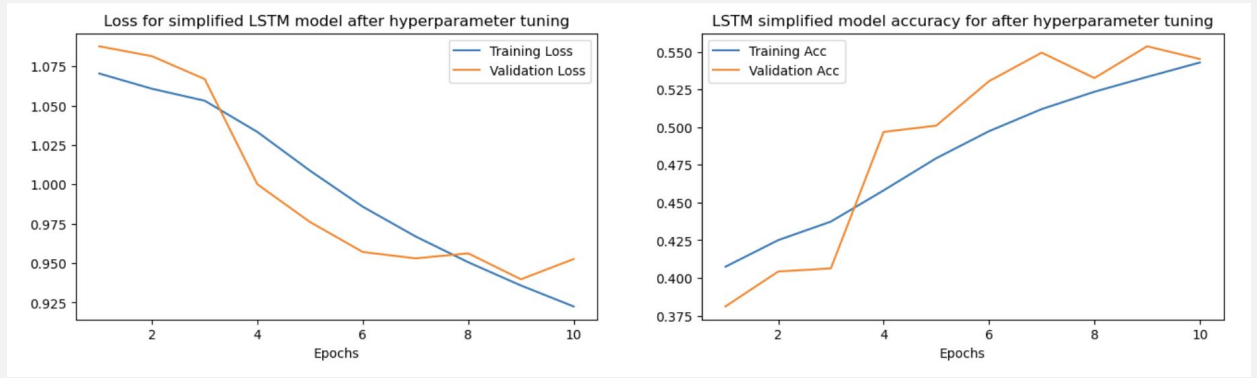
With Val/Test Sizes of 1 Day

Hyperparameters:

- Sequence length: 35 (days)
- LSTM units: 256
- LSTM layers: 1
- Val/test size: 1 day
- Learning rate: 0.001
- Batch size: 256

Final accuracies*:

- Training accuracy: 0.5429
- Validation accuracy: 0.5453
- Test accuracy: 0.54274



*Pay not much attention to the absolute accuracies, but look at the relative accuracies between them

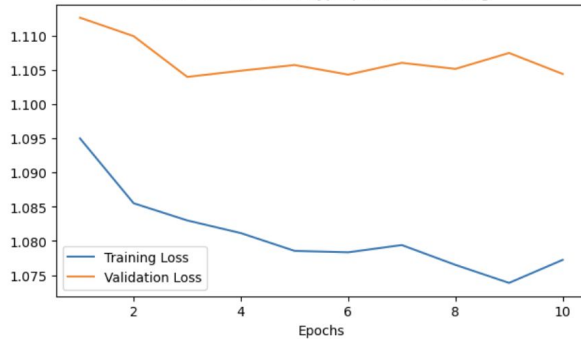
Additional Model Tests

One Stock Model

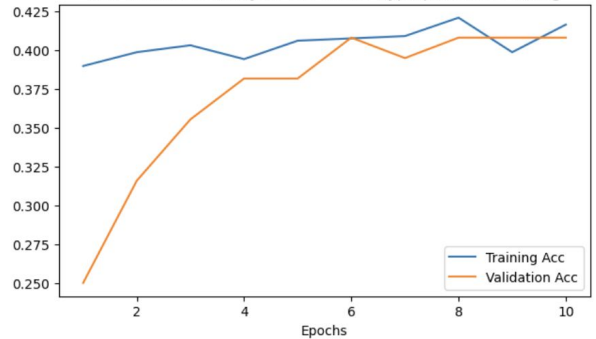
Hyperparameters:

- Sequence length: 14 (days)
- LSTM units: 256
- LSTM layers: 2
- Train/test split: 80/20
- Learning rate: 0.0001
- Batch size: 16
- Stock: AAPL

Loss for AAPL after hyperparameter tuning



LSTM model accuracy for AAPL after hyperparameter tuning



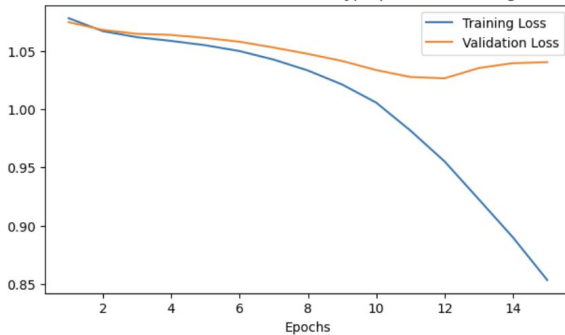
*Pay not much attention to the absolute accuracies, but look at the relative accuracies between them

One Sector Model

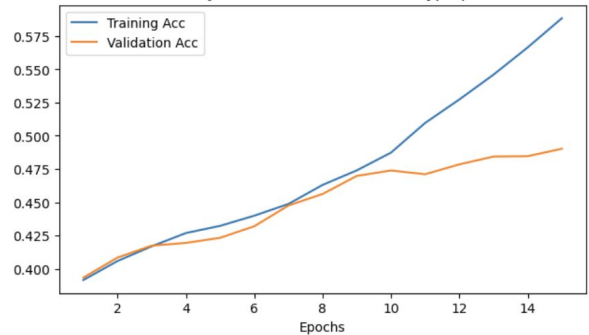
Hyperparameters:

- Sequence length: 14 (days)
- LSTM units: 256
- LSTM layers: 2
- Train/test split: 80/20
- Learning rate: 0.001
- Batch size: 128
- Sector: Information Technology

Loss for IT sector stocks after hyperparameter tuning



LSTM model accuracy for IT sector stocks after hyperparameter tuning



*Pay not much attention to the absolute accuracies, but look at the relative accuracies between them